

Sean Kim

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EDUCATION

University of California, Irvine

B.S. in Computer Science

Irvine, CA

Expected June 2027

EXPERIENCE

Software Engineer Intern

June 2025 – Sept 2025

MIADVG

Irvine, CA

- Shipped and maintained 3 production React and Next.js web applications in TypeScript with Tailwind CSS, used across products with 50,000+ combined users and features delivered in 2-week Agile sprints
- Developed Express REST APIs on Node.js connecting React frontends to PostgreSQL and MySQL, supporting OAuth 2.0 authentication and third-party integrations including Stripe and Twilio
- Automated AWS EC2 deployment with a GitHub Actions CI/CD pipeline that ran builds and a Vitest unit test suite on each merge to main, blocking failed releases while shipping 2–3 releases weekly

Software Developer Intern

Jan 2025 – May 2025

CALIT2, University of California, Irvine

Costa Mesa, CA

- Developed a Python Flask-SocketIO backend that streamed real-time detections from 5+ camera feeds to a React security dashboard over WebSockets, sustaining 100–200ms event-delivery latency
- Cut duplicate alerts by 40% on the security dashboard with a Python pipeline that deduplicated detections via a Redis TTL cache and routed them through RabbitMQ priority queues
- Fine-tuned YOLO with PyTorch transfer learning on labeled client footage, improving validation mAP by 8 points over the stock pretrained baseline powering the real-time detection pipeline

Undergraduate Research Assistant

Jan 2024 – June 2024

University of California, Irvine

Irvine, CA

- Created Python ETL pipelines with BeautifulSoup, PyPDF2, and Pandas that cleaned and normalized thousands of UCI PDF/HTML documents into training-ready text for iGPT development
- Evaluated fine-tuned GPT-3.5 responses on UCI-specific queries during early-stage iGPT development, a project that later evolved into ZotGPT for UCI students

PROJECTS

Docent | [GitHub] [Live] | *Next.js, Python, FastAPI, Pinecone, Voyage, Claude, Tavily*

Apr 2026 – Present

- Improved answers on complex questions with an agentic loop that splits each query into sub-questions, searches documents and the web, and merges them into one grounded answer
- Reduced hallucinations by hand-building the RAG pipeline to ground every answer in retrieved sources, with inline citations back to each source
- Shipped it to production as two services (Next.js + FastAPI) with per-IP rate limiting, Google sign-in, and a pytest suite, plus automated deploys on every push to main

CollabCode | [GitHub] [Live] | *React, TypeScript, Monaco, Flask-SocketIO, Redis, Postgres*

Feb 2026 – Apr 2026

- Sustained 17ms p95 sync latency with 50 concurrent clients per room by building a real-time collaborative code editor with live cursors, follow mode, and multi-file workspaces
- Enabled real-time multi-user editing by implementing the React sync layer for the Monaco editor, applying remote edits without disrupting local typing or cursor position
- Kept rooms and files intact across restarts by designing a three-layer persistence system: in-memory live state, Redis for active room state, and PostgreSQL for durable storage

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, HTML, CSS

Frameworks & Libraries: React, Next.js, Vite, Node.js, Express, Flask, FastAPI, Tailwind CSS, WebSockets, PyTorch, BeautifulSoup, PyPDF2, Pandas, Vitest, pytest

Cloud & Infra: AWS (EC2), Docker, RabbitMQ, GitHub Actions, CI/CD, Git

Databases: PostgreSQL, MySQL, Redis, Pinecone, SQLite

AI/ML: Retrieval-Augmented Generation (RAG), Vector search, Embeddings, Agentic pipelines, Fine-tuning, Transfer learning, YOLO, Anthropic SDK